EXPONENTIAL SMOOTHING

SINGLE EXPONENTIAL SMOOTHING : α KONSTAN

- Pemberian bobot pada data yang makin lama akan menurun selama eksponensial.

$$F_{t+1} = F_t + \frac{1}{N}\left(X_t - X_{t-N+1}\right)$$

$$F_{t+1} = F_t + \frac{1}{N}\left(X_t - F_t\right)$$

$$F_{t+1} = \frac{1}{N}X_t + \left(1 - \frac{1}{N}\right)F_t$$

- **N = bil positif** **N = ~** **1/N = 0**
  N = 1 **1/N= 1**
- $\alpha = 1/N$ $F_{t+1} = \alpha X_t + (1-\alpha)F_t$
- α = konstanta exp.smoothing.
  $= 0 \leq \alpha \leq 1$

SINGLE EXPONENTIAL SMOOTHING : α KONSTAN

$$F_{t+1} = \alpha X_t + (1-\alpha) F_t$$

$$\rightarrow = \alpha X_t + (1-\alpha)\left[\alpha X_{t-1} + (1-\alpha) F_{t-1}\right]$$

$$\rightarrow = \alpha X_t + \alpha(1-\alpha) X_{t-1} + (1-\alpha)^2 . F_{t-1}$$

$$\rightarrow = \alpha X_t + \alpha(1-\alpha) X_{t-1} + (1-\alpha)^2 . \left[\alpha X_{t-2} + (1-\alpha) F_{t-2}\right]$$

$$\rightarrow = \alpha X_t + \alpha(1-\alpha) X_{t-1} + \alpha(1-\alpha)^2 X_{t-2} + (1-\alpha)^3 \left[\alpha X_{t-3} + (1-\alpha) F_{t-2}\right]$$

$$\rightarrow = \alpha X_t + \alpha(1-\alpha) X_{t-1} + \alpha(1-\alpha)^2 X_{t-2} + \alpha(1-\alpha)^2 X_{t-2} + \alpha(1-\alpha)^2 X_{t-3} + (1-\alpha)^4 X_{t-4}$$

SINGLE EXPONENTIAL SMOOTHING : α BERUBAH (Adaptive Response Rate SES)

- Untuk jumlah data banyak (ratusan, ribuan)
- Rumus : =

$$F_{t+1} = \alpha_t X_t + (1 - \alpha_t) F_t$$

- Dimana :

$$\alpha_t = \left[\frac{E_{t-1}}{M_{t-1}} \right]$$

$$E_t = \beta e_t + (1 - \beta) E_{t-1}$$

$$M_t = \beta |e_t| + (1 - \beta) M_{t-1}$$

$$e_t = X_t - F_t$$

$$0 \leq \alpha_1 \beta \leq 1$$

Et = kesalahan yang dihasilkan
Mt = kesalahan absolute yang dihaluskan

SINGLE EXPONENTIAL SMOOTHING : α BERUBAH

- Contoh:

$$F_2 = X_{1F}$$
$$\alpha_2 = \alpha_3 = \alpha_4 = \beta = 0,2$$
$$E_1 = M_1 = 0$$
$$\alpha_5 = \left|\frac{E_4}{M_4}\right| = \left|\frac{-5,26}{11,38}\right| = 0,4622$$
$$F_5 = \alpha_4 X_4 + (1-\alpha_4) F_4$$
$$F_5 = 0,2(197,5) + 0,8(188,6) = 190,38$$
$$e_5 = X_5 - F_5 = 310 - 190,38 = 119,62$$
$$E_5 = \beta e_5 + (1-\beta) E_4 = 0,2(119,62) + 0,8(-5,26) = 19,716$$
$$M_5 = \beta |e_5| + (1-\beta) M_4 = 0,2(119,62) + 0,8(11,8) = 33,028$$
$$F_6 = \alpha_5 X_5 + (1-\alpha_5) F_5$$
$$F_6 = 0,4622(310) + 0,5378(190,38) = 245,67$$

.

.

.

dst

SINGLE EXPONENTIAL SMOOTHING : α BERUBAH (perbandingan nilai α)

- α = 0,2

$$X_t = 0{,}2$$

$$X_{t-1} = 0{,}16$$

$$X_{t-2} = 0{,}128$$

$$X_{t-3} = 0{,}1024$$

periode	demand	α = 0,1	α = 0,5	α = 0,9
Januari	200			
pebruari	135	200	200	200
maret	195	193.5	167.5	141.5
april	197.5	193.7	181.3	189.7
mei	310	194	189.4	196.7
juni	175	205.6	249.7	298.7
juli	155	202.6	212.3	187.4
agustus	130	197.8	183.7	158.2
september	220	191	156.8	132.8
oktober	277.5	193.9	188.4	211.3
nopember	235	202.3	233	270.9
desember		205.6	234	238.6

SINGLE EXPONENTIAL SMOOTHING : α BERUBAH

$$F_2 = \alpha X_1 + (1 - \alpha_1) F_1$$
$$= 0{,}1.200 + 0{,}9.200$$
$$= 200$$
$$F_3 = \alpha X_2 + (1 - \alpha_2) F_2$$
$$= 0{,}1.135 + 0{,}9.200$$
$$= 193{,}5$$
$$F_4 = \alpha X_3 + (1 - \alpha_3) F_3$$
$$= 0{,}1.195 + 0{,}9.193{,}5$$
$$= 193{,}7$$

SINGLE EXPONENTIAL SMOOTHING : α BERUBAH

- Semakin besar α akan memberikan smoothing yang semakin kecil → fluktuasi semakin besar
- Permasalahan yang timbul : berapa nilai α yang optimal sehingga memberikan mean square error yang kecil?

$$MSE = \sum_{i=1}^{n} \frac{e_i^2}{n}$$

$$\alpha = 0{,}1 \rightarrow MSE = 3438$$

$$\alpha = 0{,}5 \rightarrow MSE = 4347$$

$$\alpha = 0{,}9 \rightarrow MSE = 5039$$

DOUBLE EXPONENTIAL SMOOTHING (BROWN'S ONE PARAMETER LINEAR METHOD)

- DIGUNAKAN UNTUK DATA YANG MENGANDUNG TREND
- RUMUS:

$$S'_t = \alpha X_t + (1-\alpha) S'_{t-1}$$

$$S''_t = \alpha S'_t + (1-\alpha) S''_{t-1}$$

$$a_t = S'_t + (S'_t - S''_t) = 2S'_t - S''_t$$

$$b_t = \frac{\alpha}{1-\alpha}(S'_t - S''_t)$$

$$F_{t+m} = a_t + b_t m$$

- m = jumlah periode ke depan dari forecast

CONTOH

periode	demand	SES	DES	at	bt	Ft(m = 1)
1	143	143	143			
2	152	144.8	143.36	146.24	0.36	
3	161					146.6
4	139					
5	137					
6	174					
7	142					
8	141					
9	162					
10						

DOUBLE EXPONENTIAL SMOOTHING (BROWN'S ONE PARAMETER LINEAR METHOD)

- α = 0,2

$$S'_1 = \alpha X_1 + (1-\alpha) S'_0 = 0.2(143) + 0.8(143) = 143$$

$$S''_1 = \alpha S'_1 + (1-\alpha) S''_0 = 0.2(143) + 0.8(143) = 143$$

$$S'_2 = \alpha X_2 + (1-\alpha) S'_1 = 0.2(152) + 0.8(143) = 144.8$$

$$S''_2 = \alpha S'_2 + (1-\alpha) S''_1 = 0.2(144.8) + 0.8(143) = 143.36$$

$$a_2 = 2(144,8) - 143.36 = 146.24$$

$$b_2 = \frac{0,2}{0,8}(144.8 - 143.36) = 0.36$$

$$F_3 = a_2 + b_2.1 = 146.24 + 0.36(1) = 146,6$$

DOUBLE EXPONENTIAL SMOOTHING (HOLT TWO PARAMETERS LINEAR METHOD)

- MELAKUKAN SMOOTHING TERHADAP FAKTOR TREND SECARA TERPISAH
- RUMUS :

$$S_t = \alpha X_t + (1-\alpha)(S_{t-1} + b_{t-1})$$

$$b_t = \beta(S_t - S_{t-1}) + (1-\beta)b_{t-1}$$

$$F_{t+m} = S_t + b_t.m$$

Contoh :

Periode	demant	St	bt	Ft(m = 1)
1	143	143	9	
2	152	152	9	152
3	161	161	9	161
4	139	162.8	7.14	170
5	137			170.94
6	174			
7	142			
8	141			
9	162			
10				

DOUBLE EXPONENTIAL SMOOTHING (HOLT TWO PARAMETERS LINEAR METHOD)

- Nilai α = 0,2 β = 0,3

$$S_1 = X_1 = 143$$

$$b_1 = X_2 - X_1 = 152 - 143 = 9$$

$$S_2 = \alpha X_2 + (1-\alpha)(S_1 + b_1) = 0,2(152) + 0,8(143+9) = 152$$

$$b_2 = \beta(S_2 - S_1) + (1-\beta)b_1 = 0,3(152-143) + 0,7(9) = 9$$

$$F_2 = S_1 + b_1.1 = 143 + 9 = 152$$

$$S_3 = \alpha X_3 + (1-\alpha)(S_2 + b_2) = 0,2(161) + 0,8(152+9) = 161$$

$$b_3 = \beta(S_3 - S_2) + (1-\beta)b_2 = 0,3(161-152) + 0,7(9) = 9$$

$$F_3 = S_2 + b_2.1 = 152 + 9 = 161$$